Predicting Future Performance of NCAA Division I Men's Hockey Teams Using Historical and College-Specific Data

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Abstract– Despite the rapid growth of research in sports science over the past two decades, NCAA Division I men's hockey remains an underexplored area. This paper examines a critical aspect of sports analytics within Division I hockey—predicting team performance in an upcoming season. By analyzing historical data and college-specific factors, this study aims to identify key determinants of team success, contributing to the broader understanding of performance forecasting in collegiate athletics. This study aims to identify and quantify critical determinants of future team performance, addressing two central questions: (1) which historical and institutional features best predict upcoming season success, and (2) how roster continuity impacts prediction accuracy. Leveraging historical performance data (2013–2023), institutional rankings, and coaching tenure information, this research employs and compares several machine learning techniques, including Random Forest, AdaBoost, and Neural Network models. Results indicate that aggregated plus/minus (+/–) of returning players overwhelmingly emerges as the most influential predictor, followed by game-winning goals (GWG). The AdaBoost model demonstrated superior predictive performance, achieving a mean absolute error (MAE) of 0.0977 and an adjusted R^2 of approximately 0.770, outperforming Random Forest and Neural Network models. Importantly, the study found minimal predictive value from factors like institutional prestige, geographical location, and simple roster continuity metrics. Practically, these insights suggest NCAA teams could benefit from prioritizing player retention strategies that emphasize individual plus/minus metrics rather than broader roster stability or institutional factors. Despite constraints in

historical data—particularly in advanced player metrics and pandemic-era gaps—this study offers vital insights into the key drivers of a team's future performance.

1 INTRODUCTION

NCAA Division I men's hockey is a competitive collegiate sport played at many large colleges and universities across the United States. The game is played on ice, with each team consisting of five skaters and one goaltender. A standard match consists of three 20-minute periods, with the objective of scoring more goals than the opposing team. Throughout a season, teams compete in numerous games, which can be categorized into exhibition games, tournament games, and regular-season conference games.

This study focuses specifically on regular-season conference games for several reasons. First, each team plays a relatively similar number of conference games per season, typically ranging from 22 to 28. Second, these games are highly competitive, as they directly impact rankings and postseason qualification, unlike exhibition games, where teams may experiment with new strategies or rotate players to minimize injury risks. Lastly, focusing on conference games ensures consistency in the analysis, as teams compete against opponents within their league, reducing variability caused by mismatches that can occur in exhibition games.

Each conference game results in one of three possible outcomes: a win, where a team scores more goals than its opponent in regulation or overtime; a tie, where both teams score the same number of goals in regulation and remain tied after overtime; or a loss, where a team scores fewer goals than its opponent in regulation or overtime. For the purpose of this study, a point system will be used to quantify these outcomes: teams earn two points for a win, one point for a tie, and zero points for a loss.

Unlike professional hockey leagues, NCAA Division I hockey teams experience continuous roster turnover due to eligibility restrictions. Under NCAA rules, student-athletes have five years of eligibility from the time they enroll as full-time students at any post-secondary institution, with a maximum of four years of competition. This rapid turnover rate makes recruiting and coaching

staff vital for maintaining a consistent level of team performance. Unlike players, coaching staff are not subject to strict eligibility limits, allowing them to provide stability and continuity over multiple seasons. As a result, the ability to effectively recruit, develop, and integrate new talent is essential for sustained success in NCAA Division I men's hockey.

This study examines the influence of various factors on predicting team performance in the upcoming season. Key variables include the past performance of individual players and the team, the tenure and historical success of coaching staff, the desirability of the college for prospective athletes, and the geographical location of the institution. By analyzing these and other relevant factors, this research aims to identify the most significant determinants of team success in NCAA Division I men's hockey, contributing to the broader field of sports analytics in collegiate hockey.

To develop predictive models, this study will explore multiple approaches. A random forest model will be implemented as a primary method due to its robustness in handling high-dimensional data and its ability to generate interpretable reports on factor importance. Additionally, a neural network model will be tested to capture complex, non-linear relationships that a random forest model may struggle to identify. Lastly, an Adaboost model will be tested as it was found to have the highest performance of Ice Hockey prediction models in a review of sports prediction performance (Bunker & Susnjak, 2022). By comparing these approaches, this study seeks to determine the most effective methodology for forecasting team performance.

2 LITERATURE REVIEW

Research specifically predicting NCAA hockey outcomes is limited, yet several related studies offer valuable insights. The following review synthesizes findings from various sports prediction studies, highlighting key takeaways that inform this research.

2.1 The Relationship Between Model Complexity and Predictive Accuracy

Bunker and Susnjak (2019) conducted a comprehensive review of machine learning applications in team sport prediction spanning 1996 to 2019. Their

analysis revealed a critical insight: model complexity does not necessarily correlate with enhanced predictive accuracy. Specifically, their examination of English Premier League prediction models demonstrated that models utilizing fewer than 10 features often achieved comparable performance to those incorporating up to 320 features.

Furthermore, the review highlighted a historical reliance on artificial neural networks (ANNs) as the primary predictive tool, with earlier studies indicating that "23% of the papers considered solely ANNs as their predictive model" (Bunker & Susnjak, 2019). However, the evidence presented suggests that ANNs do not consistently outperform alternative machine learning algorithms (Bunker & Susnjak, 2019). Consequently, a more diversified approach, exploring a range of models such as random forests and AdaBoost, as previously mentioned, may yield more robust and reliable predictive outcomes than a singular focus on neural network methodologies.

2.2 Penalty Minutes in Hockey Predictions

Buttrey (2015) developed a method for predicting the outcome of National Hockey League (NHL) games using a model that combines goal scoring and yielding with penalty commission. The study utilized four years of NHL game data (2010-2011 through 2014-2015) to predict win probabilities. As previously noted, Buttrey's (2015) study reveals insights into the nature of penalty minutes, suggesting that the concept of “coincident penalties” might make penalty minutes less predictive than they initially appear. This observation is particularly relevant for NCAA hockey, where datasets may lack detailed penalty context, potentially diminishing the weight of penalty minutes as a feature.

2.3 Historical Data Consideration

Erazo (2023) investigated the effect of historical data volume on predictive accuracy in basketball using a hybrid simulation approach with an NBA case study. The study aimed to understand the quality of predictions using limited variables and the optimal amount of historical data needed for accuracy. Results indicated that an optimal range of data quantity exists, suggesting that more data is not always better and that the selection of data and variables is crucial. This

finding supports the idea that there may be a “sweet spot” for data input in sports prediction models, a principle that could guide the selection of historical data for NCAA hockey analysis.

2.4 Application of Wins Above Replacement (WAR) Principles

The Wins Above Replacement (WAR) metric, widely utilized across various sports, quantifies an individual's contribution to team performance relative to a defined replacement-level player. Specifically, WAR represents the incremental wins attributed to a player beyond the expected output of a readily available substitute (Baumer, Jensen, & Matthews, 2015). While a standardized WAR model is not currently established for NCAA men's hockey, the methodological frameworks employed in other sports offer valuable insights.

Baumer, Jensen, and Matthews (2015) introduced openWAR, an open-source system designed to evaluate comprehensive player performance in Major League Baseball. This framework underscores the significance of contextualizing player contributions across diverse game facets, including hitting, pitching, baserunning, and fielding. Similarly, Eager (2020) developed a WAR model for the National Football League, leveraging Pro Football Focus (PFF) player grades to elucidate player value and positional variations.

The adaptation of such methodologies to NCAA men's hockey presents an opportunity to refine feature weighting in predictive models. For instance, the principles of WAR suggest a potentially higher weighting for 'goals against' in the evaluation of defensive players and 'goals for' in the assessment of offensive players. The successful implementation of WAR in baseball and football demonstrates the potential for developing comprehensive metrics that account for positional responsibilities, thereby informing the nuanced weighting of player statistics within NCAA hockey prediction models.

2.5 Consideration of Additional Variables

The introduction of this study highlights the importance of factors beyond traditional on-ice statistics in predicting NCAA hockey performance, including coaching staff tenure and success, college desirability for athletes, and geographical location. While direct research on these specific factors in sports

prediction might be limited, the impact of coaching is implicitly acknowledged in the need to evaluate team performance and strategy. Furthermore, the continuous roster turnover in NCAA hockey due to eligibility restrictions, as detailed in the introduction, underscores the potential influence of coaching staff stability and recruiting effectiveness on sustained team success. The study by Bunker and Susnjak (2019) reviewed various features used in sports prediction, suggesting that a comprehensive approach considering both player and team-level factors is often employed. While college desirability and geographical location are less commonly explored in quantitative prediction models, they represent potential areas for future research in collegiate sports analytics.

By synthesizing these insights from related sports prediction research, this literature review provides a valuable foundation for the current study on predicting NCAA men's hockey outcomes. The findings on model complexity, the nuanced nature of penalty minutes, the importance of optimal historical data ranges, and the potential for adapting comprehensive metrics like WAR will guide the development and evaluation of the predictive models explored in this research.

3 RESEARCH QUESTION

This analysis focuses on the ability to predict how an NCAA Division I men's hockey team will perform in an upcoming season after knowing what the roster for the next year will be with a high degree of confidence, specifically examining the seasons between 2013 and 2024. The following research questions will guide the course of the study:

1. *What are the most significant features of team performance in an upcoming season between 2013 and 2024, considering variables such as past team and player performance, college desirability, coach tenure, and geographical latitude?*

2. *How does roster continuity from one season to the next affect the predictive accuracy of models forecasting team performance during the 2013–2024 NCAA Division I men's hockey seasons?*

4 DATA & METHODOLOGY

This study employs a quantitative methodology to analyze and predict NCAA Division I men's hockey team performance for the seasons spanning 2013-2014 through 2023-2024. The process involves comprehensive data collection, preprocessing, exploratory data analysis, and the application and evaluation of machine learning models.

4.1 Data Collection

The data collection phase focused on compiling a comprehensive dataset for each team across the specified seasons.

Institutional Factors: Each institution's ranking was obtained from U.S. News & World Report. This ranking includes two components: the category (National University, National Liberal Arts College, or Regional University/College) and the specific rank within that category. The category is treated as an ordinal variable (coded 0, 1, and 2 for National, National Liberal Arts, and Regional classifications, respectively). The specific rank is utilized as a numerical variable. It is noted that multiple institutions across different regions may share the same numerical rank within the Regional categories. Additionally, the latitude of each college's location was incorporated as a potential proxy for geographical influences on hockey program strength.

Coaching Data: The complete history of head coaches for each team prior to and during the study period was sourced from collegehockeynews.com. Coach tenure is defined as the number of consecutive seasons an individual served as head coach for that specific institution, resetting to one upon any break or change in leadership. The historical regular-season points per game (PPG) achieved by each coach at the institution was tracked for every season they served as head coach. For coaches in their inaugural season at an institution, the team's PPG from the immediately preceding season was used as an initial estimate of coaching baseline performance.

Player and Goalie Statistics: Statistics from the preceding season were aggregated for all returning players (identified as described in Section 4.4). For skaters, this included games played (GP), goals (G), assists (A), penalty minutes (PIM), shots on goal (SOG), game-winning goals (GWG), faceoffs won (FOW), and faceoffs

lost (FOL). For goalies, aggregated statistics included goals against (GA), saves (SV), shutouts (SO), and minutes played (Min). For both goalies and players names are also gathered to assist in tracking specific players over multiple seasons. This aggregation of statistics from returning players serves to quantify the retained talent base for each team, forming a key input for predicting points per game (PPG) in the subsequent season. It is acknowledged that statistics from incoming freshmen are not included in this predictive feature set.

Upon completion of data collection, the dataset was structured with one row per team per season, encompassing institutional attributes, coaching factors, and aggregated returning player/goalie statistics, along with engineered features described below.

4.2 Data Preprocessing

Before analysis and modeling, the collected data underwent preprocessing to ensure data quality, consistency, and suitability for machine learning algorithms.

Handling Missing Data: The datasets gathered were generally complete. All universities included in the analysis possessed rankings, and the core hockey statistics were available. A notable exception pertains to the 2020-2021 season, significantly impacted by the COVID-19 pandemic. Teams that did not participate or had substantially incomplete schedules during the 2020-2021 season were excluded from the analysis for that year due to the absence of reliable performance data. Consequently, these teams were also excluded from the 2021-2022 season analysis, as the models require prior season performance data which was unavailable or unreliable from 2020-2021.

Data Type Conversion: Data types were validated and corrected where necessary. Specifically, the 'Season' column, originally formatted textually (e.g., '2013-14'), was transformed to represent the starting year of the season numerically (e.g., 2013). The goalie 'Min' (minutes played) column, initially in a 'MM:SS' format, was converted to a decimal representation of total minutes to facilitate numerical analysis.

Consistency Checks: To ensure consistent team identification across disparate data sources, a unique identifier ('Rk') originating from the collegehockeynews

dataset was adopted. For datasets lacking this identifier but containing team names, a matching process was implemented. Team names were standardized (converted to lowercase, stripped of punctuation, special characters, and whitespace) and joined with a reference key dataset containing 'Rk' values to assign the correct identifier consistently.

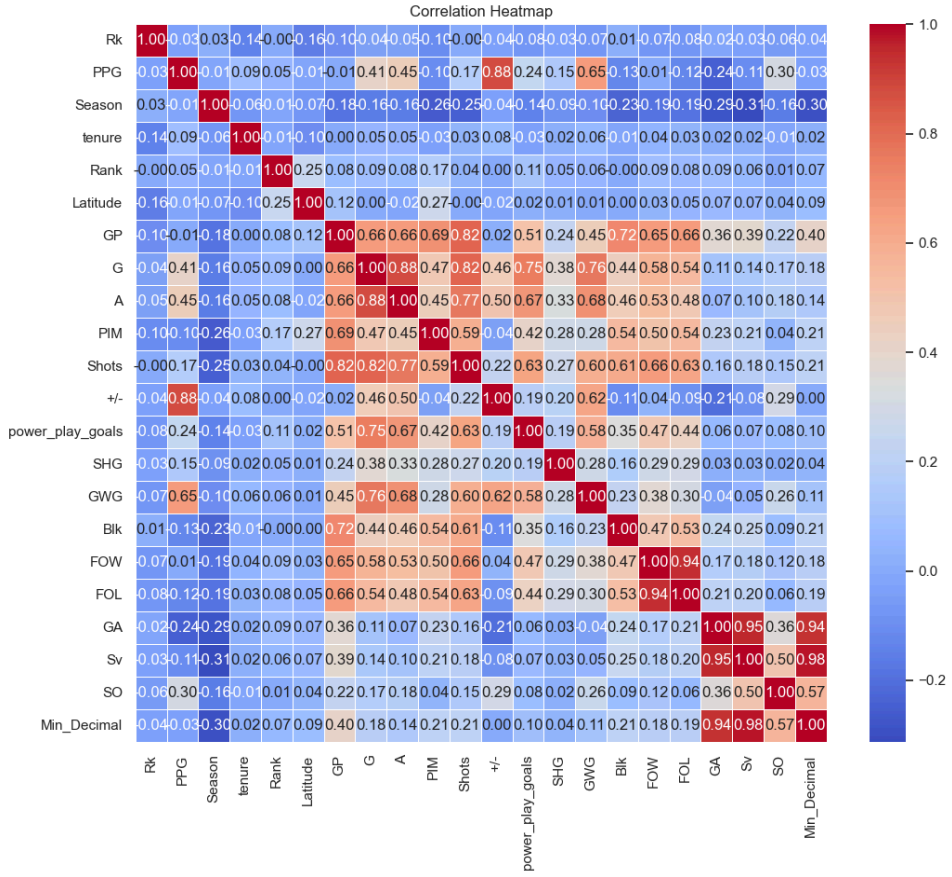
4.3 Exploratory Data Analysis (EDA)

To gain insights into the dataset's characteristics, identify patterns and relationships, validate assumptions, and inform modeling strategies, Exploratory Data Analysis (EDA) was conducted. The primary objectives were to understand the distributions of individual variables, visualize relationships between potential features and the target variable (Points Per Game - PPG), assess correlations (including potential multicollinearity), identify any residual data quality issues, and gather preliminary evidence related to the research questions and hypotheses.

Statistic	Goals	Shots	Goals Against	Goalie Minutes
Count	556	556	556	556
Mean	43.54	467.25	35.26	808.37
Std. Dev.	14.34	118.9	27.63	610.04
Min	0	19	0	0
25th Percentile	34	385.75	5	92.86
Median (50%)	42	455.5	38	905.29
75th Percentile	52	539.25	57	1350.4
Max	95	874	103	1726.2

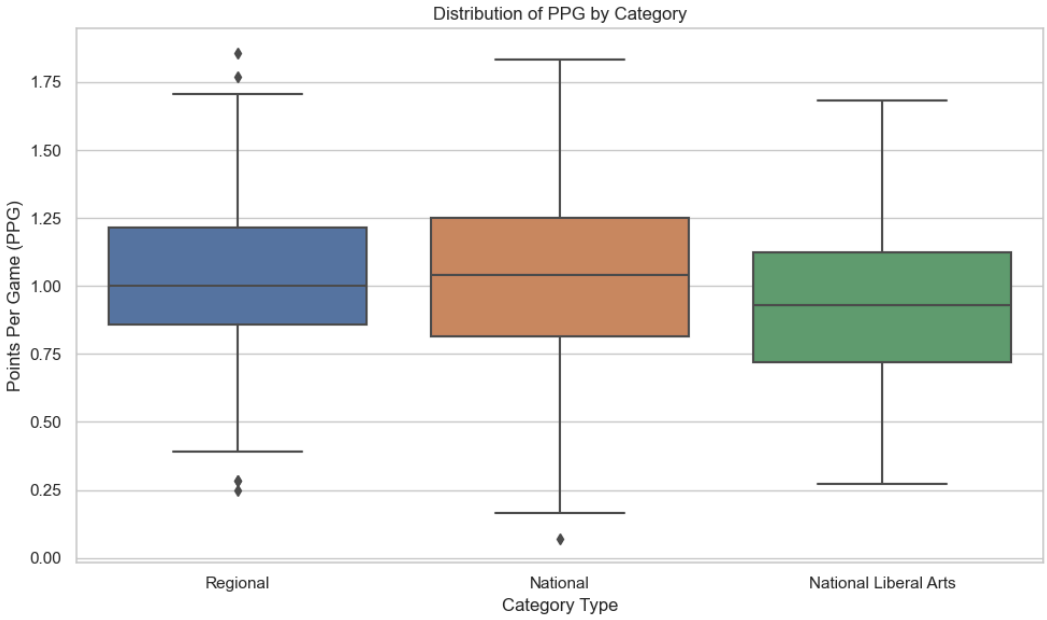
Missing Values	0	0	0	0

The table above shows the summary statistics of a select few important columns for players and goalies.



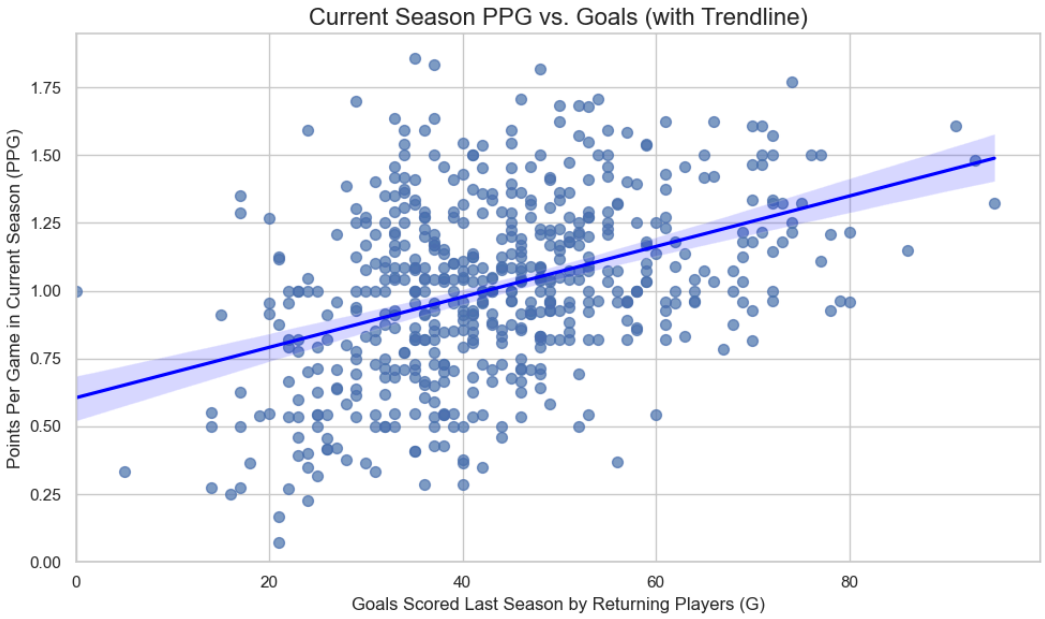
Examination of the correlation heatmap revealed several key relationships. Among the aggregated returning skater statistics, two features exhibiting strong positive correlations with the target variable (PPG) were the aggregated plus/minus statistic (+/-) and the aggregated count of game-winning goals (GWG). The significance of the aggregated +/- statistic is intuitive, as it provides a composite measure reflecting a player's contribution during even-strength play relative to opposition scoring while on the ice. Another reason for +/- having such an outsized impact could be due to it being the only feature of those measured that can be either increased or decreased by any particular player, unlike goals, shots, or many of the other statistics where the lowest value any individual player can have assigned to them is 0, with +/- a player can have either

a positive or negative value with no constraints on how positive or negative the value can be. Conversely, the strong correlation observed for aggregated GWG was less anticipated. While game-winning goals are crucial in specific game contexts, it was initially hypothesized that all goals scored by returning players might contribute similarly to overall future team PPG potential, rather than GWG having disproportionate predictive value. Furthermore, the analysis highlighted potential multicollinearity among key aggregated goalie statistics. Specifically, strong positive correlations were observed between returning goalie Goals Against (GA), Saves (SV), and Minutes Played (Min). This collinearity can pose challenges for certain modeling techniques, particularly those sensitive to redundant features. High multicollinearity can inflate the variance of coefficient estimates in some models (e.g., linear regression), making it difficult to isolate the independent effect of each correlated variable and potentially impacting model interpretability and stability, although tree-based models like Random Forest are generally less sensitive to it.

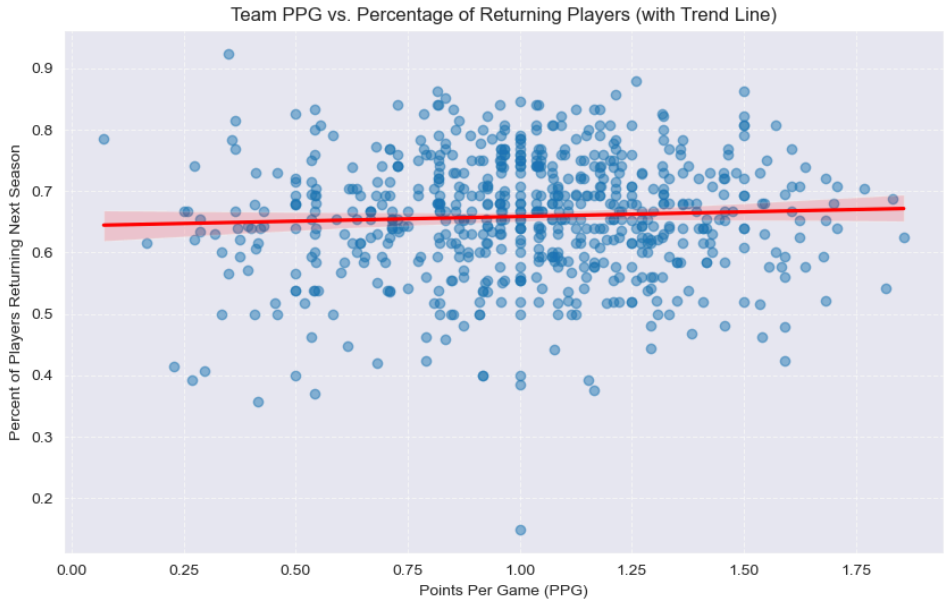


Analysis of PPG distribution across US News ranking categories using box plots (referencing the provided figure) revealed nuanced differences. National universities exhibited the highest median PPG (approximately 1.05), slightly exceeding Regional institutions (median ~0.98). National Liberal Arts colleges showed a marginally lower median performance (median ~0.93) compared to the

other two categories. In terms of variability, National universities displayed the narrowest interquartile range (IQR), suggesting greater consistency among the middle 50% of teams in this category, whereas Regional and National Liberal Arts institutions showed wider IQRs, indicating greater performance dispersion within their respective groups. While the median performance levels across categories show some proximity, the differences in distribution shape and variability suggest that ranking category may still hold some relevance for team performance characteristics.



A scatter plot comparing the total goals scored in the previous season by returning players against the team's PPG in the current season revealed a distinct positive linear trend, as illustrated by the superimposed regression line. This indicates that teams retaining players who collectively scored more goals in the prior season tend to achieve higher points per game in the subsequent season. This relationship aligns with intuition, as a stronger returning offensive base is expected to contribute positively to future team success, translating goals scored into points earned through wins and ties. While the positive trend is evident, the plot also shows considerable variance around the regression line, suggesting that returning goals alone do not fully determine future PPG and other factors play substantial roles.



The scatter plot illustrates the relationship between a team's Points Per Game (PPG) and the percentage of players returning from one season to the next. The nearly flat trend line, accompanied by minimal slope and a broad dispersion of data points around the regression line, suggests only a weak correlation. This indicates that the sheer percentage of returning players is not strongly predictive of future performance. Rather, it is likely that the quality and individual contributions of these returning players carry greater significance in determining subsequent season success.

4.4 Feature Engineering

This study incorporates several engineered features derived from the collected data to potentially enhance predictive capability.

The primary engineered feature is the target variable itself, Points Per Game (PPG). PPG is calculated for each team-season based on conference game outcomes (assigning 2 points for a win, 1 point for a tie, and 0 points for a loss), divided by the total number of conference games played by the team in that season. This provides a normalized measure of regular-season conference performance.

Coach Tenure (as defined in section 4.1) is utilized directly as a feature representing coaching stability.

Coach PPG is the average PPG of a coach for the previous seasons of the coach's tenure with a team.

The concept of Roster Continuity is applied during the data aggregation phase rather than as a separate engineered feature variable. It is derived by checking the prior season's roster for names that are also present on the roster for the current season (represented conceptually as a 0/1 flag per player). This player-level identification is then used to determine which players' statistics are included when aggregating goals, assists, and all other statistics as described in Section 4.1. Therefore, the aggregated statistics inherently represent the contributions of returning players only.

Aside from the specifically calculated target variable (PPG) and the directly used Coach Tenure, the primary input features representing retained on-ice talent are the aggregated player and goalie statistics resulting from this filtered process.

4.5 Methodology: Predictive Modeling

To address the research questions, predictive models were developed using the preprocessed and explored data. The primary goal is to predict Points Per Game (PPG) for NCAA Division I men's hockey teams in an upcoming season based on the collected historical, institutional, and aggregated returning player data.

This study will explore and compare the performance of multiple machine learning approaches selected for their suitability to this type of regression task:

Random Forest: Chosen for its robustness in handling potentially high-dimensional data, its inherent ability to capture non-linear relationships, and its capability to provide interpretable reports on feature importance, which directly addresses the research question regarding significant features.

Neural Network: Included to explore potentially complex, non-linear interactions between variables that might not be fully captured by tree-based methods. The Neural Network model will utilize a multi-layer perceptron (MLP) architecture.

The specific configuration, including the number of hidden layers and nodes per layer, will be determined through systematic experimentation and validation performance.

AdaBoost (Adaptive Boosting): Selected based on its documented success in sports prediction contexts (Bunker & Susnjak, 2022) and its mechanism of focusing sequentially on harder-to-predict instances, potentially yielding high predictive accuracy, particularly when applied with decision tree base learners (AdaBoost Regressor).

Training and Testing Split: The dataset will be partitioned chronologically to simulate a realistic forecasting scenario. Data from the 2013-2014 through 2021-2022 seasons (corresponding to start years 2013 through 2021) will constitute the training set. Data from the 2022-2023 and 2023-2024 seasons (start years 2022 and 2023) will be reserved as the hold-out test set for final model evaluation.

Model Evaluation Metrics: Model performance will be evaluated primarily using the Mean Absolute Error (MAE). MAE provides an easily interpretable measure of average prediction error in the original units of the target variable (PPG). Given the naturally bounded nature of PPG (typically between 0 and 2), MAE is considered a robust metric less sensitive to the potential impact of outliers compared to squared-error metrics. Additionally, the Adjusted R-squared (Adjusted R^2) metric will be reported to provide insight into the proportion of variance in the target variable explained by the models, adjusted for the number of features.

Feature Importance Assessment: To identify the most significant features of team performance, feature importance will be assessed for each model type:

For Random Forest, importance scores will be derived using permutation importance on the held-out test set. This method measures the decrease in model score when a single feature's values are randomly shuffled, providing a reliable estimate of feature contribution to predictive performance. Mean Decrease in Impurity (Gini importance) may also be examined as a secondary measure derived from the training process.

For AdaBoost, feature importance will be assessed based on the feature_importances_ attribute, which typically reflects the total reduction of the loss function brought by splits on that feature across all boosting rounds or how often a feature was selected.

For the Neural Network (MLP), interpreting feature importance is less direct. Permutation importance will again be employed as a model-agnostic approach.

The importance rankings from the different models will be compared to identify features consistently ranked as highly influential across diverse modeling techniques.

By systematically developing, training, and evaluating these diverse modeling approaches, this study aims to identify the most effective methodology and the key determining factors for forecasting team performance in NCAA Division I men's hockey within the specified timeframe and data constraints.

5 HYPOTHESIS

Null Hypothesis (H_0): The percentage of freshmen through juniors on the previous year's roster is not a primary predictor of performance, and the additional factors (previous season performance, academic grades, school desirability ranking, coach tenure, etc..) do not significantly predict performance.

Alternative Hypothesis (H_1): The percentage of freshmen through juniors on the previous year's roster is the primary predictor of performance, with additional significant features including the team's previous season performance, academic grades, and the school's desirability ranking.

6 THEORY

NCAA Division I Men’s hockey team performance is influenced by a dynamic interplay of recruitment, historical momentum, player attributes, and institutional prestige:

- **Youthful Recruitment:** A higher number of freshmen through juniors on the previous year's roster signals an influx of emerging talent that can drive innovation and adaptation in team strategy.

- **Historical Momentum:** The team's performance in the previous season reflects established team dynamics, coaching effectiveness, and the overall competitiveness of the program.

- **Institutional Prestige:** School desirability, as ranked by US News, indicates the institution's overall appeal and resources, which can enhance recruitment efforts and provide better support systems making these schools more desirable to top recruits.

Together, these elements create a framework where the infusion of new talent is paramount, while past performance, academic prowess, and school reputation collectively shape the team's future success.

7 RESULTS AND DISCUSSION

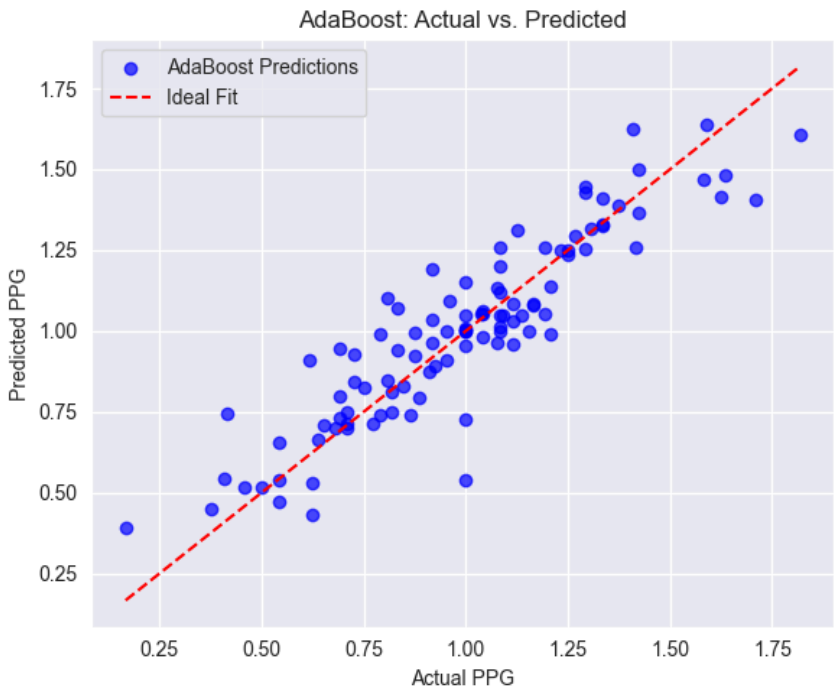
A total of 454 team-season observations (seasons 2013–2021) comprised the training set, and 102 observations (seasons 2022–2023) formed the test set.

7.1 AdaBoost Regressor

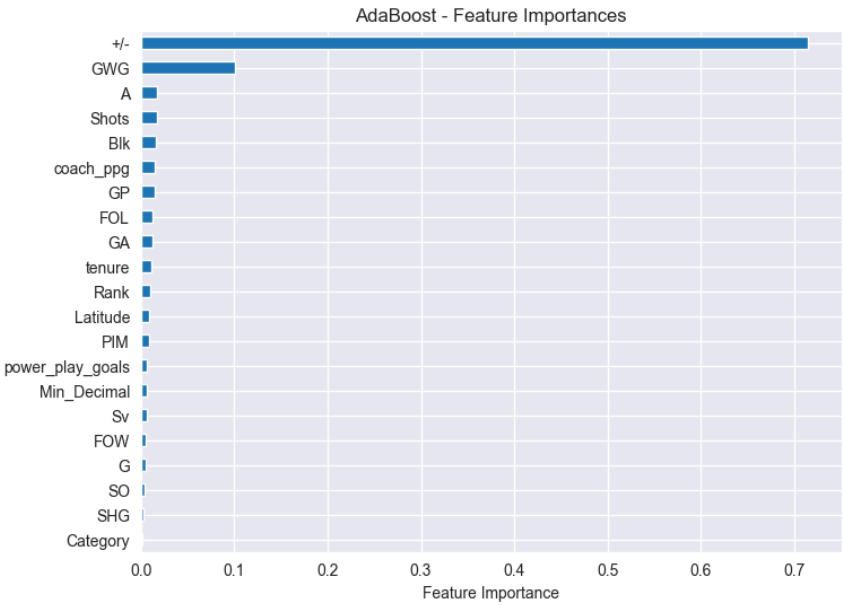
- Best parameters: n_estimators=250, learning_rate=2, decision-tree min_samples_split=10, max_depth=11

Test performance

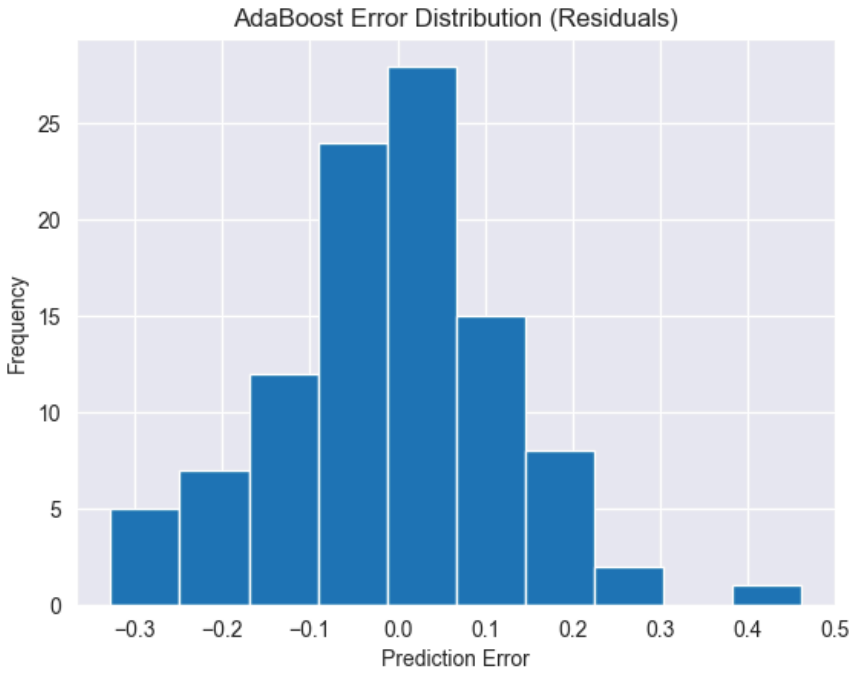
- MAE = 0.0977
- $R^2 = 0.818$
- Adjusted $R^2 = 0.770$



Feature importance reveals that aggregated plus/minus (+/–) is by far the strongest feature—≈7× the importance of game-winning goals (GWG), which itself is ≈5× the next feature. These align with the top correlations seen in Section 4.3.



Prediction errors are centered near zero, and the model accurately forecasts high and mid-range PPG values, with ≈61 % of predictions within ±0.1 PPG of actual.

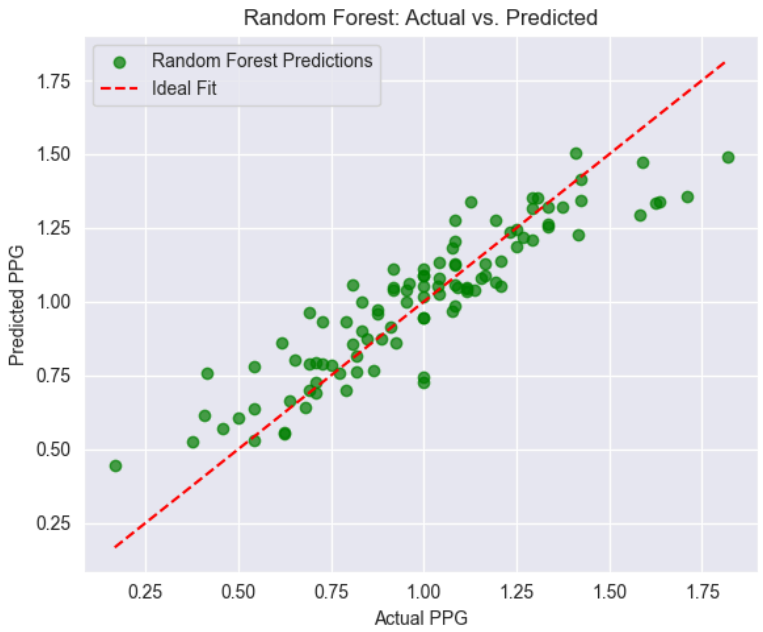


7.2 Random Forest Regressor

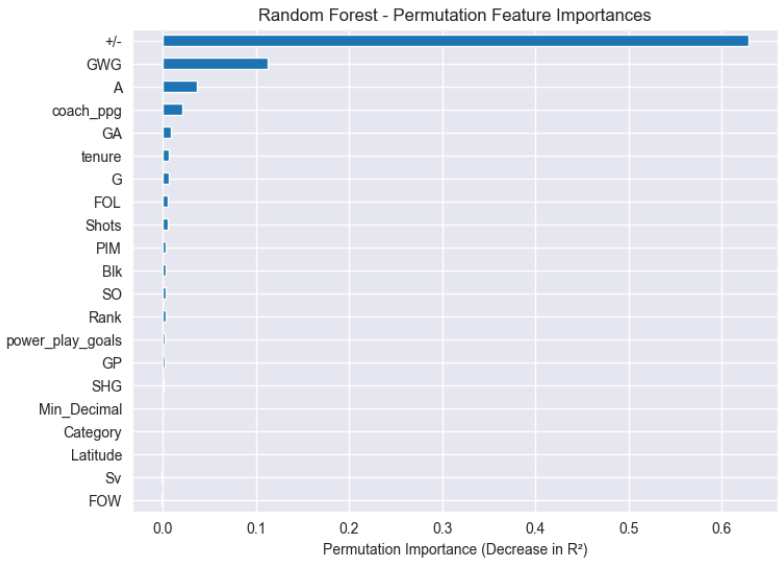
- Best parameters: n_estimators=200, min_samples_split=3, min_samples_leaf=1, max_features='sqrt', max_depth=40, bootstrap=False

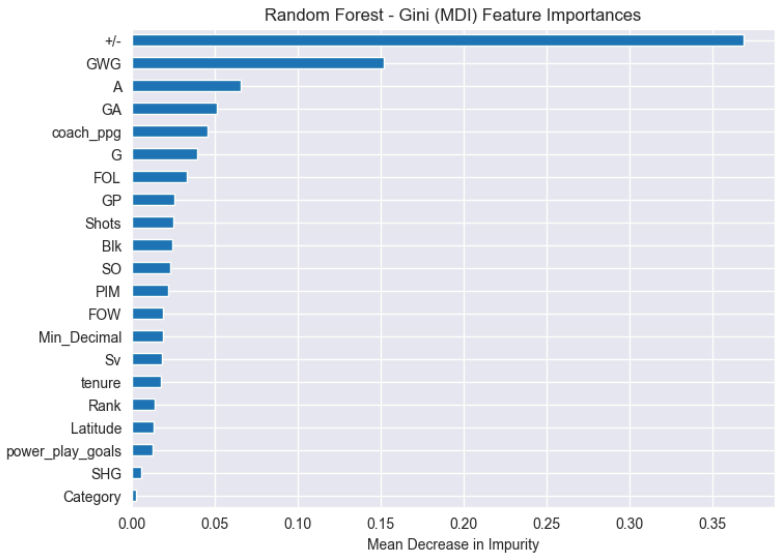
Test performance

- MAE = 0.1034
- R^2 = 0.812
- Adjusted R^2 = 0.763

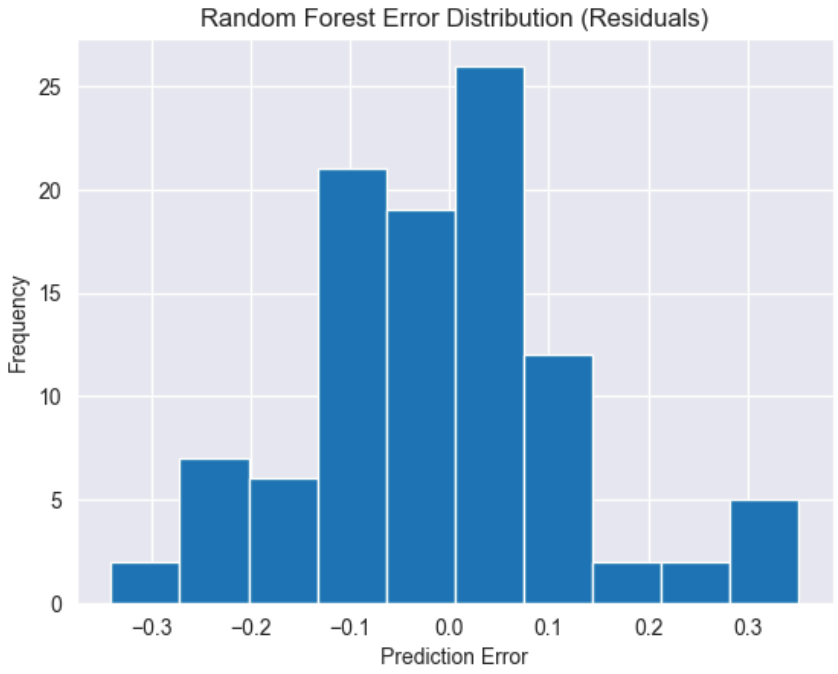


Feature importance again highlights +/– and GWG, but assigns more weight than AdaBoost does to assists (A), goals against (GA), and coach's historical PPG (coach_ppg).





Prediction behavior shows strong accuracy around average PPG (~1.0), with a solid ≈65% of predictions within ±0.1 PPG of actual, but underestimates extreme high-performance seasons—predictions plateau near 1.5 PPG.

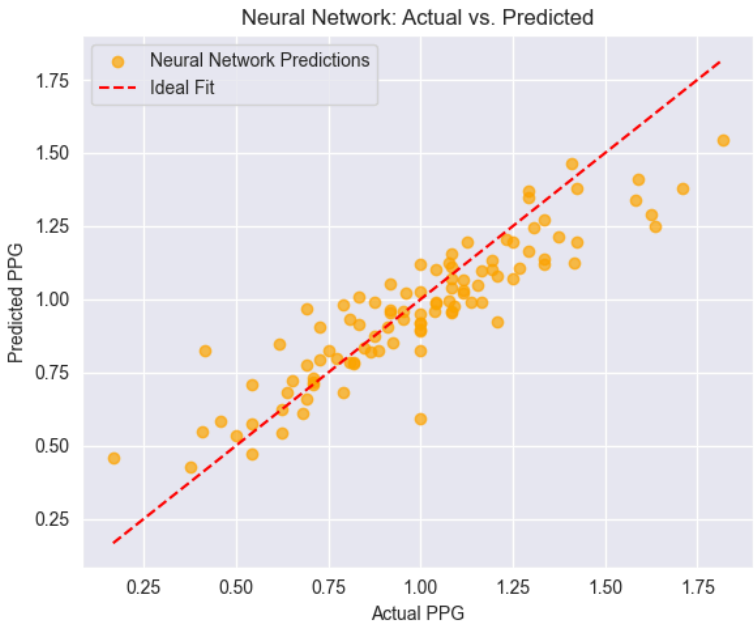


7.3 Neural Network (MLP)

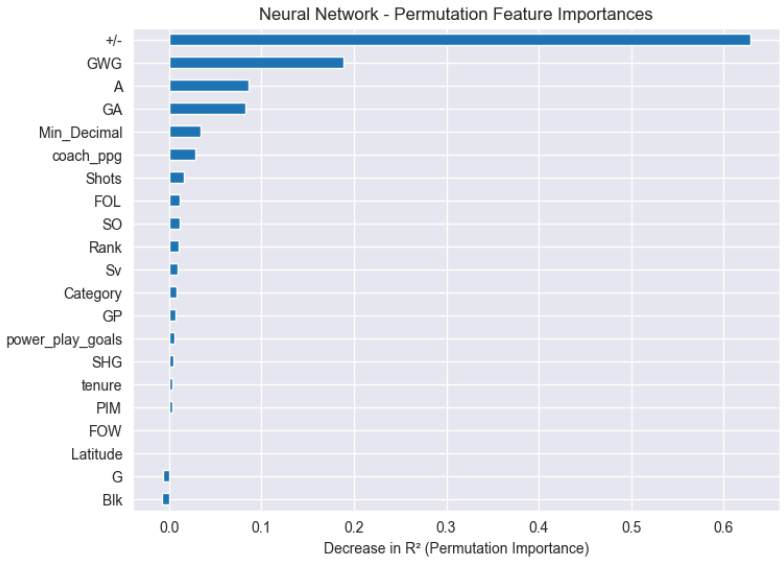
- Architecture: three hidden layers (128, 64, 32 ReLU units) plus a linear output; early stopping at 84/200 epochs

Test performance

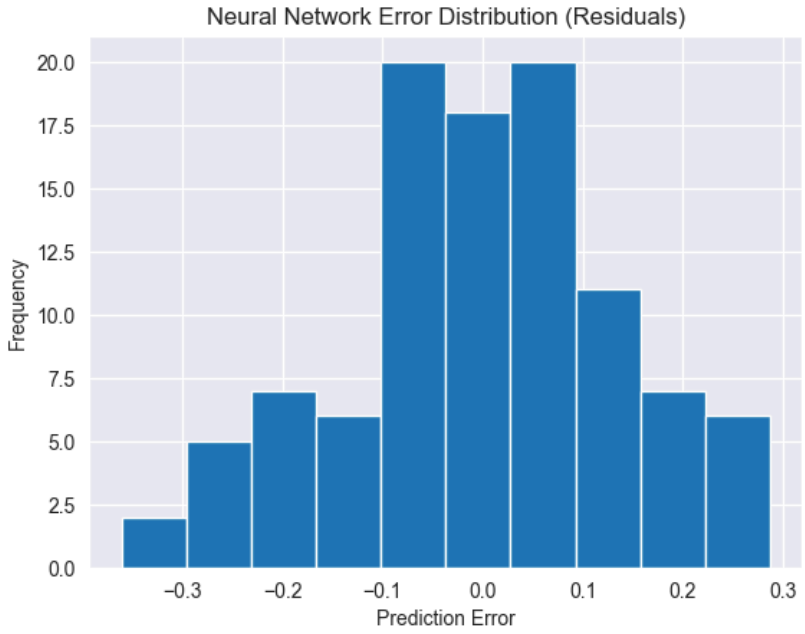
- MAE = 0.1061
- R^2 = 0.806
- Adjusted R^2 = 0.729



Permutation importance mirrors the Random Forest pattern: +/– and GWG lead, with notable contributions from A and GA.



Prediction behavior shows accuracy for mid-range PPG, similar underprediction of high outliers as seen in the Random Forest.

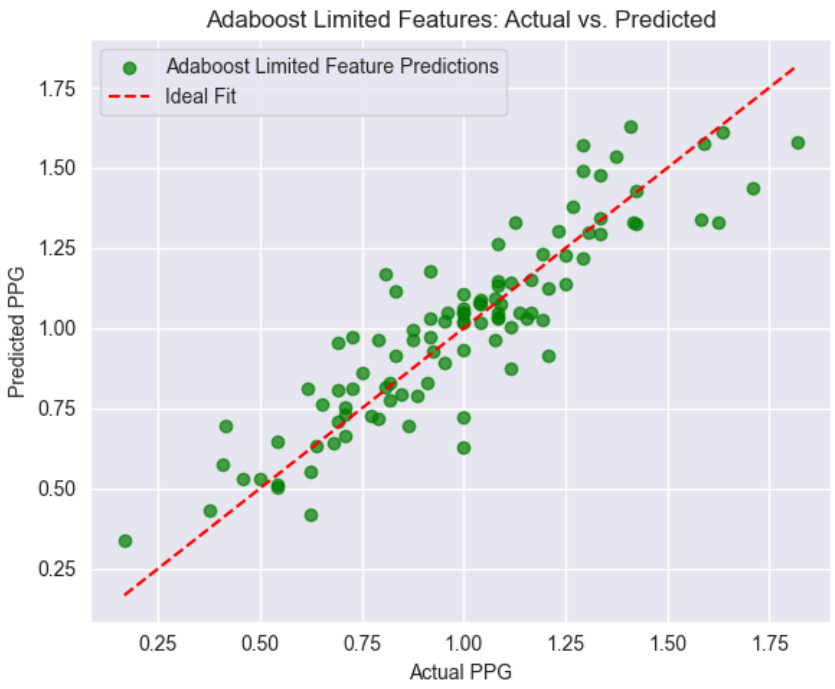


7.4 Reduced-Feature AdaBoost

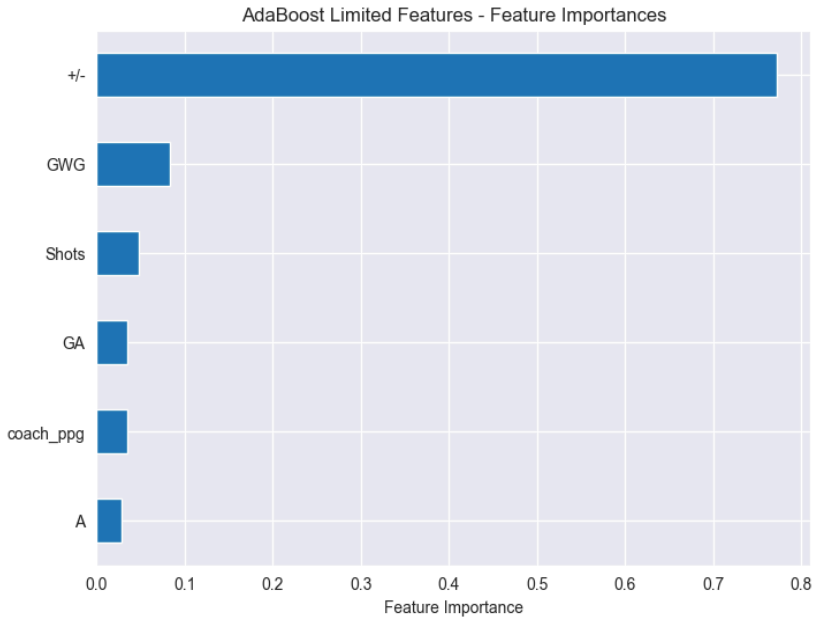
- Features used: +/–, GWG, coach_ppg, A, Shots, GA
- Best parameters: n_estimators=100, learning_rate=1.5, min_samples_split=20, max_depth=5

Test performance

- MAE = 0.1063
- R^2 = 0.798
- Adjusted R^2 = 0.785



Reducing to six features raises MAE and lowers R^2 slightly, but increases adjusted R^2—indicating a more parsimonious model with similar explanatory power. Feature importance is more even, though +/– remains dominant.



7.5 Model Comparison

Model	MAE	R^2	Adjusted R^2
Adaboost	0.097686	0.811982	0.762627
Random Forest	0.103382	0.817954	0.770167
Neural Network	0.107442	0.810684	0.760989
Limited-Features	0.106305	0.797747	0.784973

All four models consistently identify +/– and GWG as the strongest features. AdaBoost (full feature set) achieves the best MAE and R^2, while the reduced-feature AdaBoost balances simplicity and fit. The tendency of tree-based methods and the MLP to underpredict extreme high-PPG seasons suggests further refinement—perhaps additional non-linear features or ensemble techniques—may be needed to better capture those outliers.

8 CONCLUSION

This study investigated the feasibility of predicting the future performance of NCAA Division I men's hockey teams using features derived from institutional characteristics, coaching effectiveness, and returning player statistics. The primary objective was to provide colleges and coaching staff with actionable insights into factors that reliably indicate future team success, enabling informed decision-making regarding roster composition and recruitment strategies.

The analysis clearly established that the aggregated plus/minus (+/–) statistic of returning players is the most critical predictor of future team performance. Additionally, game-winning goals (GWG) were found to significantly influence predictions, demonstrating a stronger predictive impact compared to total goals scored (G), which surprisingly offered limited predictive value. Other features showing modest yet meaningful contributions included total shots, goals against (GA), historical points per game achieved by the coach (coach_ppg), and assists (A). A model utilizing these select features demonstrated an adjusted R-squared value comparable or superior to more comprehensive models while exhibiting only a slightly higher mean absolute error (MAE).

The study specifically addressed two null hypotheses. The first—that the percentage of freshmen through juniors returning to the team is not a primary predictor of performance—could not be disproved, as data showed no significant correlation between the percentage of returning players and subsequent team success. Conversely, the second hypothesis—that additional features (previous season performance, academic rankings, institutional desirability, coach tenure) do not significantly impact future performance—was disproved, with models explaining approximately 80% of the variance in future performance. However, it is essential to emphasize that nearly all predictive power derived from just a few predictors, particularly the plus/minus statistic and game-winning goals. The analysis also revealed minimal predictive impact from factors such as institutional location, academic rankings, coach tenure, and various detailed game statistics like blocked shots, faceoff outcomes, and goalie saves.

In terms of model performance, AdaBoost consistently outperformed other modeling approaches in both R-squared values and MAE, establishing it as the most suitable predictive model in this context. While all models agreed on the top two predictors (+/– and GWG), variability in secondary predictor importance informed the strategic selection of predictors for a streamlined limited-feature model.

It is important to acknowledge several methodological limitations. Predictor selection was constrained by consistent availability across the study period (2013–2023). Recent seasons introduced additional advanced metrics such as time-on-ice, shot-location data, and CHIP (College Hockey Individual Point value, similar to Wins Above Replacement), which were excluded due to their absence in earlier datasets. Future research incorporating these advanced metrics could enhance predictive accuracy further. Additionally, the study used 2024 college rankings uniformly across all study years due to historical data constraints, potentially introducing minor inaccuracies. Furthermore, disruptions caused by the COVID-19 pandemic necessitated the exclusion of some team-season observations due to incomplete data.

Practically, these findings suggest a strategic shift for NCAA hockey programs, emphasizing player evaluation based primarily on plus/minus statistics and game-winning goals rather than traditional metrics like total goals, faceoff

outcomes, or blocked shots. Resources dedicated to player recruitment and retention should prioritize individuals with strong plus/minus metrics over maintaining specific coaching personnel. Additionally, institutions can confidently conclude that geographic location or academic ranking alone does not significantly disadvantage their athletic competitiveness.

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